

Ng Kai Zheng

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 1 year of experience building enterprise-scale systems on Azure. Modernized legacy platforms serving **7,000+** internal users and architected an AI-driven workflow engine to streamline complex business processes. Malaysian citizen, ready to relocate to Singapore.

CORE COMPETENCIES

- .NET 8+ / C# Backend
- React & TypeScript Frontend
- Micro-frontends (Module Federation)
- Clean Architecture & Domain-Driven Design
- CQRS / MediatR & REST APIs
- AI Agent Orchestration & RAG
- Azure Cloud (Service Bus, Key Vault, App Insights)
- ABAC & RBAC & OAuth 2.0

WORK EXPERIENCE

Etiqa Insurance and Takaful Sdn Bhd

Nov 2025 - Present

.NET Developer — Malaysia

- Rapid System Modernization:** Accelerated delivery timelines by **50%**, consistently completing complex system revamp modules in 1 week vs. the projected 2-week lead time.
- AI Agent Orchestration:** Engineered a multi-agent system (Analysis → Plan → Implement → Review) managing context window constraints through specialized sub-agents, resulting in a **60%** boost in team productivity.
- Dynamic Workflow Engine:** Architected a module-agnostic, dynamic approval engine supporting complex routing (Parallel/Sequential/Percentage) and **100% auditability** for recruitment and internal modules.
- Micro-frontend (MFE) Migration:** Spearheaded the transition to an MFE architecture using Module Federation, enabling independent deployment cycles and ensuring **zero downtime** for unaffected modules.
- AI-Driven Recruitment:** Integrated AI analysis services to automate candidate screening, providing match scores, "red flag" detection, and summaries to streamline the hiring pipeline.
- System Architecture & Design:** Collaborated with the Customer Data Hub team to define domain boundaries and ERDs, establishing a single source of truth for employee data
- Granular Authorization:** Implemented a comprehensive Attribute-Based Access Control (ABAC) system across React and .NET for fine-grained user permission management.
- Internal HRMS (Full SDLC):** Delivered recruitment/onboarding and OKR modules in ~2 months as a core developer on a 4-person team. Serves 7,000+ internal users (Maybank group).
- API Gateway (Reverse Proxy):** Built a .NET-based gateway repository handling routing, token-bucket rate limiting, and TLS termination.
- Observability:** Configured Azure Application Insights for centralized logging and monitoring of application health.
- AI Spec-Driven Development:** Established an AI-assisted spec-driven workflow (OpenSpec + Obsidian) that keeps specs, coding standards, and requirements continuously up to date — making it easy to pick up work in a new session with full context.

AIO SYNERGY SDN BHD

June 2024 - Feb 2025

Internship Game Developer — Malaysia

- Developed server-integrated game features using TypeScript, emphasizing the creation of reusable and optimized code for various game mechanics.

- Spearheaded the backend integration for a Scratch Card game, implementing REST API calls for secure data handling, user progress tracking, and client-server communication, leading to a successful deployment.
- Collaborated in an Agile team using Git for version control, participating in the development lifecycle from debugging to testing and deployment of web-based applications.

PROJECTS

Pet Appointment System

Full-Stack Cloud-Native Web App

C#, .NET, SQL Server, Azure, AWS

Developed a full-stack pet appointment management system using Clean Architecture, implementing secure authentication and authorization using both JWT and OAuth 2.0 (Google). Utilized CQRS (MediatR) for separation of concerns; built a responsive React frontend integrated with a robust .NET backend API; designed optimized SQL Server schemas with triggers; wrote comprehensive unit tests; automated appointment reminders via Quartz.NET with WhatsApp integration; deployed to Azure App Services and AWS EC2/CloudFront with DNS via Route 53.

AI-Powered RAG Chat Demo

RAG Application

.NET, Azure AI Services

Developed a .NET Web API implementing Retrieval-Augmented Generation (RAG) using Azure AI services (embeddings and chat orchestration) to answer user queries contextually from a custom blog-post corpus.

Blockchain Multiplayer Game

Final Year Project

Unity, Azure PlayFab, ThirdWeb, Polygon, Photon Fusion

Architected and implemented a scalable backend using Azure PlayFab for a Unity multiplayer game, managing player data, authentication, and server-side logic. Integrated blockchain by connecting to ThirdWeb smart contracts on the Polygon network, developing secure server-side routines for NFT ownership verification and transactions. Utilized Photon Fusion for authoritative server networking.

EDUCATION

Bachelor of Computer Science, Graphics and Multimedia Software

Oct 2021 - July 2025

Universiti Teknologi Malaysia

CGPA: 4.0

AWARDS & HONORS

Dean's List — All Semesters (1-8) — Universiti Teknologi Malaysia

SKILLS

Languages: C#, TypeScript, JavaScript, SQL, C++, Java

Backend: .NET (Core/8+), EF Core, MediatR (CQRS), Dapper, REST APIs, Microservices, Quartz.NET

Frontend: React, Micro-frontends (MFE), Module Federation, Tailwind CSS, Redux

AI & Automation: AI Agent Orchestration, Prompt Engineering, RAG, Multi-Agent Workflows

Architecture & Security: Clean Architecture,DDD, ABAC, JWT, OAuth 2.0, Result Pattern

Cloud & DevOps: Azure (App Services, Storage, Service Bus, Key Vault, App Insights), AWS (EC2, CloudFront, Route 53), Docker, Git, Jira